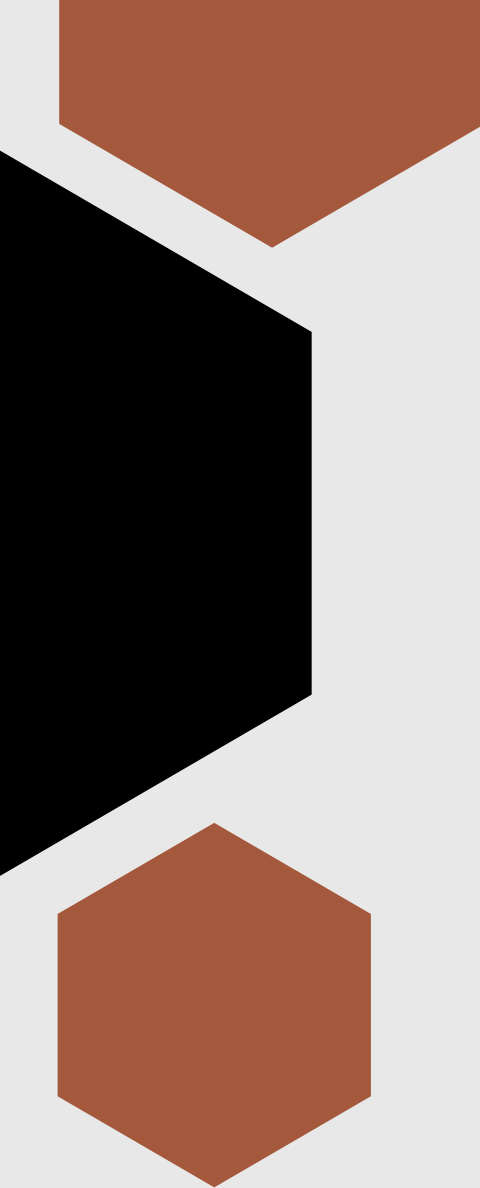


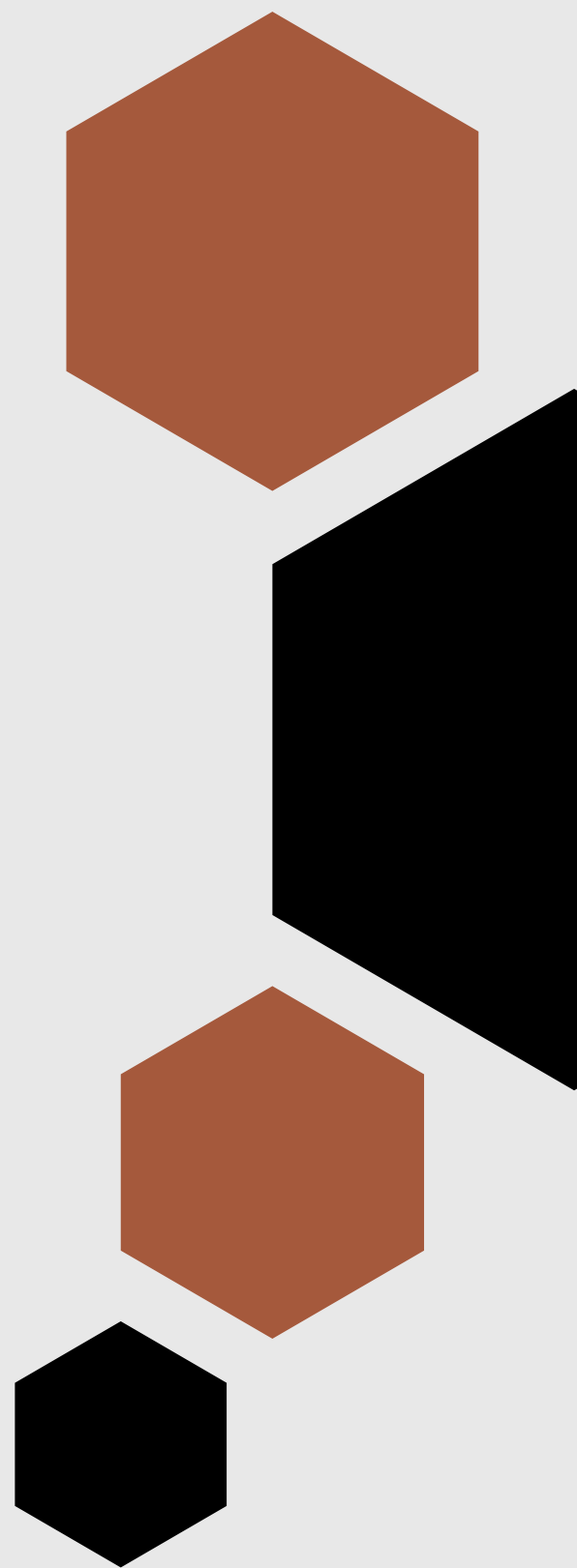


REAL ESTATE PRICE EVALUTION USING LINEAR REGRESSION

DATA ANALYSIS AND MODEL PREDICTION

RISHIKA V
MANASA J
NUR HALIZA H

DATE: 19.07.2025



PROJECT OVERVIEW

This project focuses on predicting real estate prices using Linear Regression. The data includes features such as house age, MRT station distance, nearby convenience stores.

WORK FLOW:

- Data Understanding & Preprocessing
- Data Visualization(EDA)
- Model Training
- Model Evaluation & Results

DATASET DESCRIPTION

TEXT :

Source : UCI Machine learning Repository

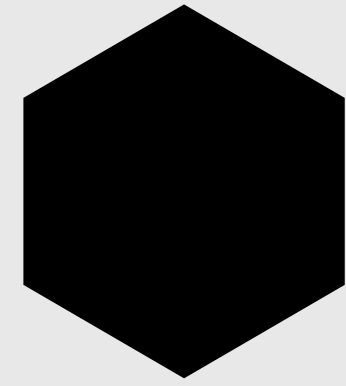
Features:

- Transaction Date
- House age
- Distance to Nearest MRT
- Number of Convenience Stores
- Latitude & Longitude

Target Variables:

House price per unit area

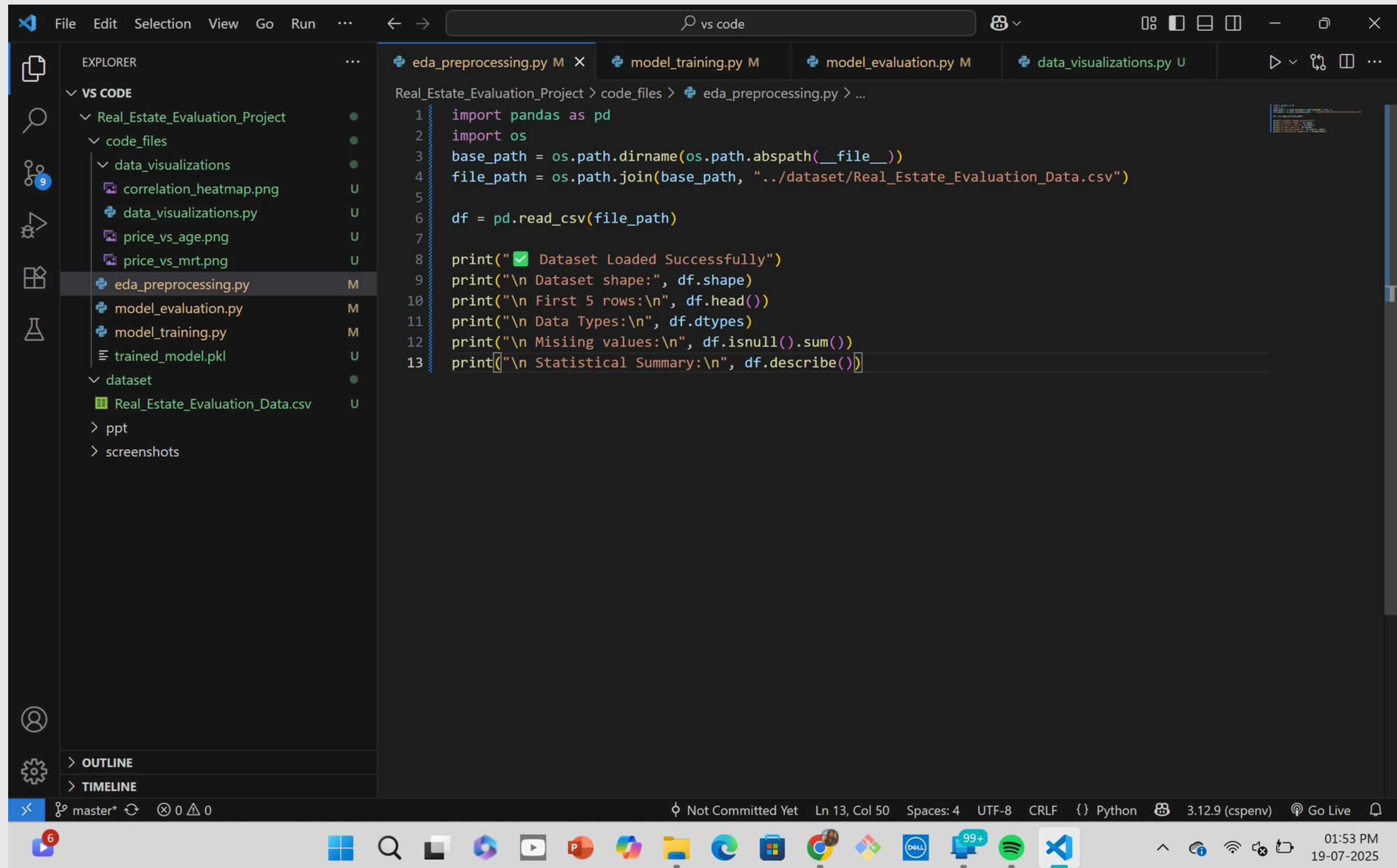
DATA PREPROCESSING



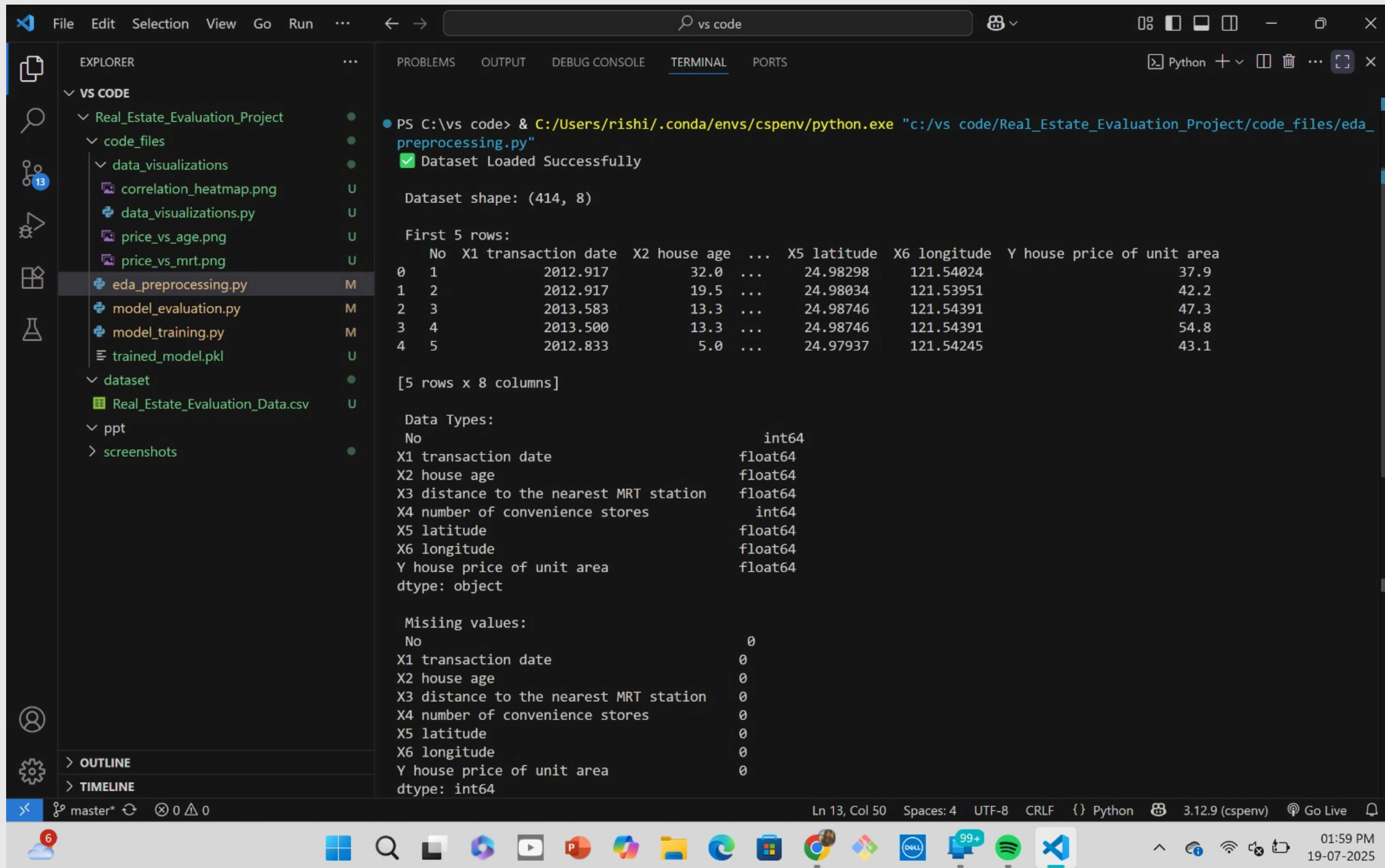
TEXT :

- Dropped ID column("NO")
- Cleaned and selected features:
 - X1 to X6 (Numerical)
 - Y as target (House price)
- Checked for null values (None found)
- Normalization not applied (simple regression)

EDA PREPROCESSING



EDA PREPROCESSING-OUTPUT



The screenshot shows the VS Code interface with the Explorer panel on the left displaying the project structure. The main editor area shows the output of a Python script executed in the terminal. The output includes the dataset shape, the first 5 rows of data, and the data types for each column.

```
PS C:\vs code> & C:/Users/rishi/.conda/envs/cspenv/python.exe "c:/vs code/Real_Estate_Evaluation_Project/code_files/eda_preprocessing.py"
[✓] Dataset Loaded Successfully

Dataset shape: (414, 8)

First 5 rows:
   No  X1 transaction date  X2 house age  ...  X5 latitude  X6 longitude  Y house price of unit area
0    1          2012.917         32.0  ...    24.98298    121.54024                37.9
1    2          2012.917         19.5  ...    24.98034    121.53951                42.2
2    3          2013.583         13.3  ...    24.98746    121.54391                47.3
3    4          2013.500         13.3  ...    24.98746    121.54391                54.8
4    5          2012.833          5.0  ...    24.97937    121.54245                43.1

[5 rows x 8 columns]

Data Types:
No                                int64
X1 transaction date              float64
X2 house age                    float64
X3 distance to the nearest MRT station float64
X4 number of convenience stores  int64
X5 latitude                    float64
X6 longitude                    float64
Y house price of unit area      float64
dtype: object

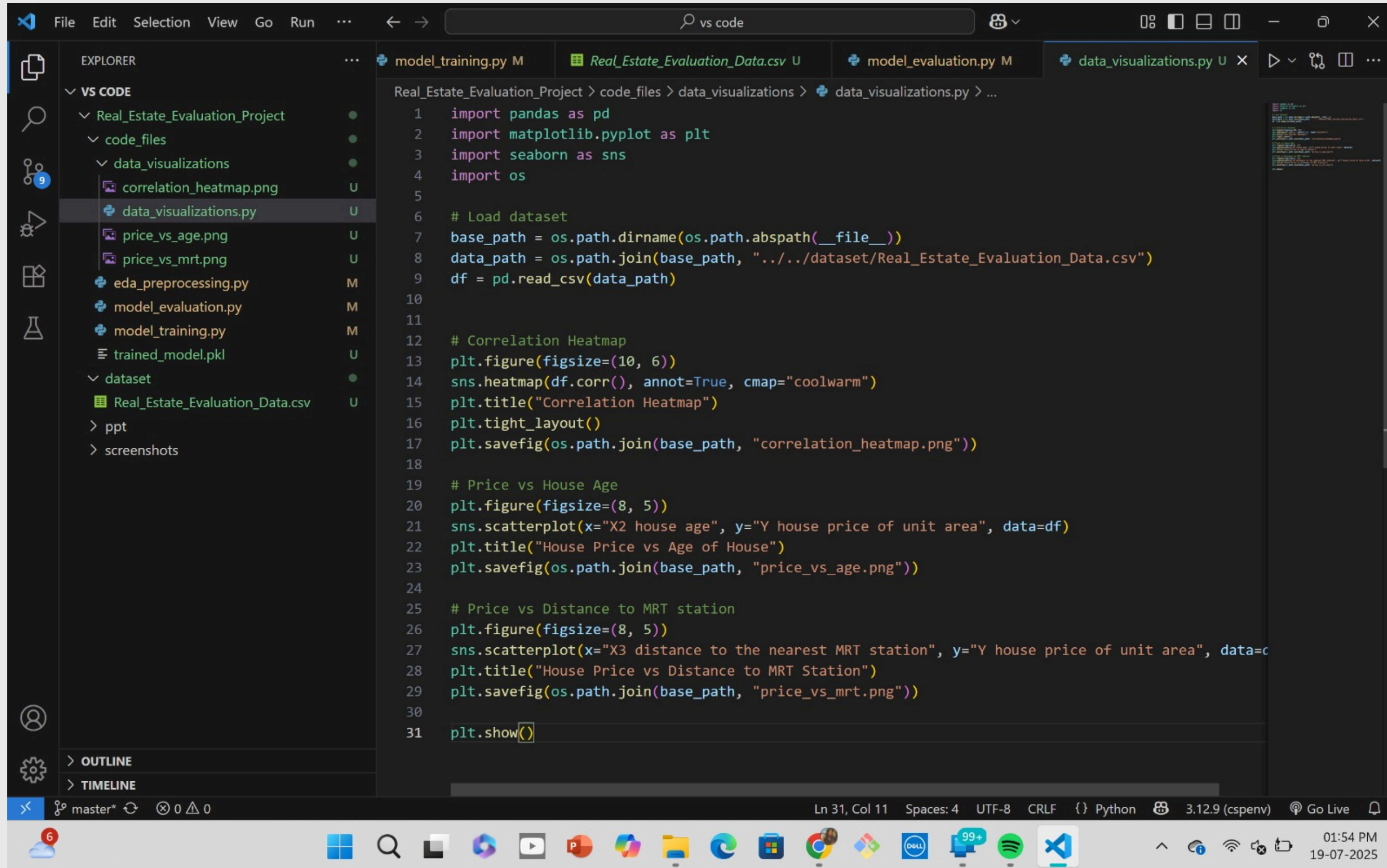
Missing values:
No                                0
X1 transaction date              0
X2 house age                    0
X3 distance to the nearest MRT station 0
X4 number of convenience stores  0
X5 latitude                    0
X6 longitude                    0
Y house price of unit area      0
dtype: int64
```

The Explorer panel shows the following structure:

- VS CODE
 - Real_Estate_Evaluation_Project
 - code_files
 - data_visualizations
 - correlation_heatmap.png
 - data_visualizations.py
 - price_vs_age.png
 - price_vs_mrt.png
 - eda_preprocessing.py (selected)
 - model_evaluation.py
 - model_training.py
 - trained_model.pkl
 - dataset
 - Real_Estate_Evaluation_Data.csv
 - ppt
 - screenshots

The status bar at the bottom indicates the current file is at Line 13, Column 50, with 4 spaces, UTF-8 encoding, and CRLF line endings. The Python interpreter is set to 3.12.9 (cspenv).

DATA VISUALIZATION



The image shows a screenshot of the Visual Studio Code (VS Code) editor interface. The main window displays a Python script named `data_visualizations.py` located within the `Real_Estate_Evaluation_Project` directory. The script is currently open in the editor, showing the following code:

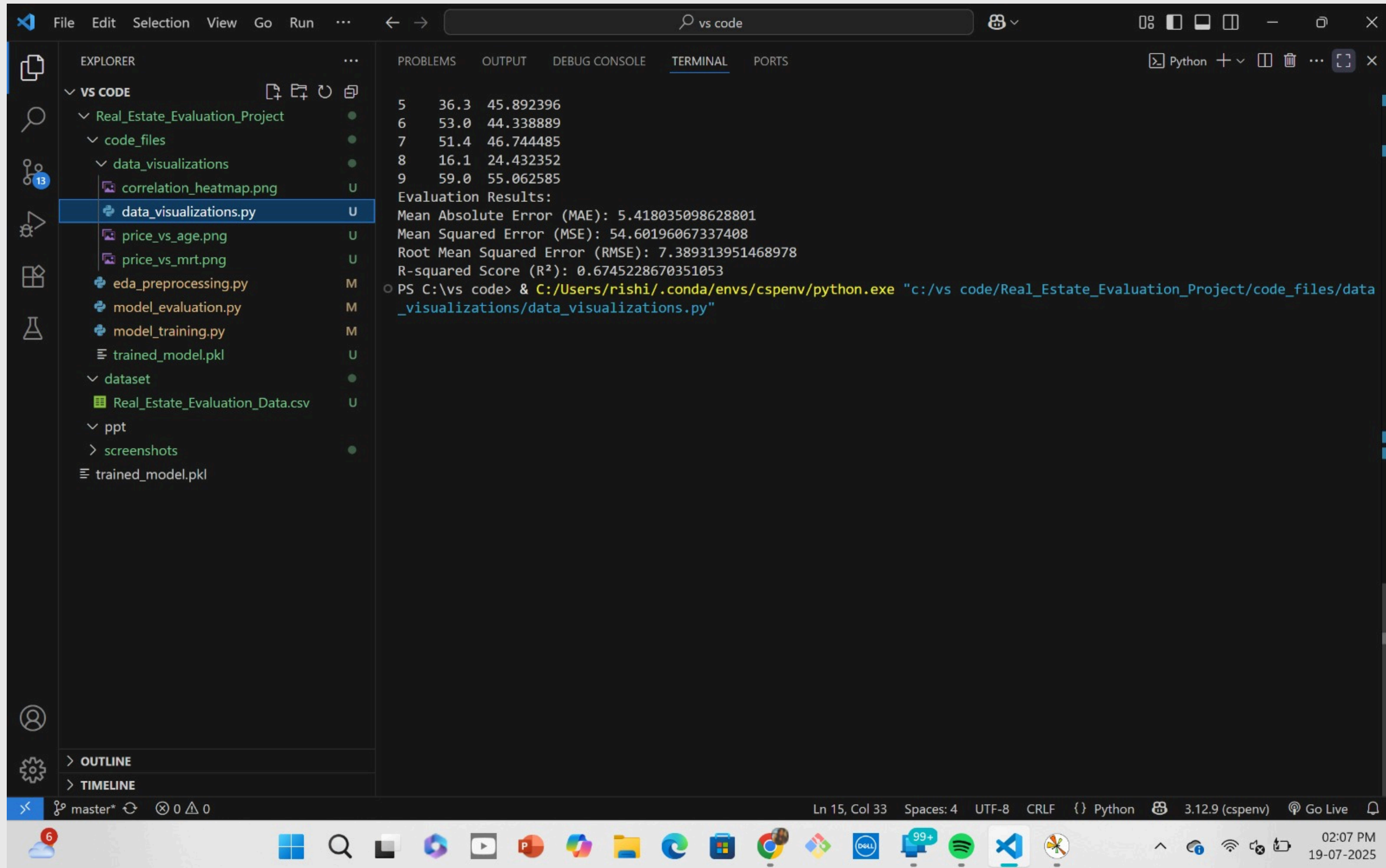
```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4 import os
5
6 # Load dataset
7 base_path = os.path.dirname(os.path.abspath(__file__))
8 data_path = os.path.join(base_path, "../../dataset/Real_Estate_Evaluation_Data.csv")
9 df = pd.read_csv(data_path)
10
11
12 # Correlation Heatmap
13 plt.figure(figsize=(10, 6))
14 sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
15 plt.title("Correlation Heatmap")
16 plt.tight_layout()
17 plt.savefig(os.path.join(base_path, "correlation_heatmap.png"))
18
19 # Price vs House Age
20 plt.figure(figsize=(8, 5))
21 sns.scatterplot(x="X2 house age", y="Y house price of unit area", data=df)
22 plt.title("House Price vs Age of House")
23 plt.savefig(os.path.join(base_path, "price_vs_age.png"))
24
25 # Price vs Distance to MRT station
26 plt.figure(figsize=(8, 5))
27 sns.scatterplot(x="X3 distance to the nearest MRT station", y="Y house price of unit area", data=df)
28 plt.title("House Price vs Distance to MRT Station")
29 plt.savefig(os.path.join(base_path, "price_vs_mrt.png"))
30
31 plt.show()
```

The left sidebar shows the Explorer view with the project structure:

- VS CODE
 - Real_Estate_Evaluation_Project
 - code_files
 - data_visualizations
 - correlation_heatmap.png
 - data_visualizations.py
 - price_vs_age.png
 - price_vs_mrt.png
 - eda_preprocessing.py
 - model_evaluation.py
 - model_training.py
 - trained_model.pkl
 - dataset
 - Real_Estate_Evaluation_Data.csv
 - ppt
 - screenshots

The bottom status bar indicates the current file is `data_visualizations.py`, line 31, column 11, with 4 spaces, UTF-8 encoding, and CRLF line endings. The Python interpreter is set to `3.12.9 (cspenv)`. The system tray at the bottom shows the time as 01:54 PM on 19-07-2025.

DATA VISUALIZATION-OUTPUT

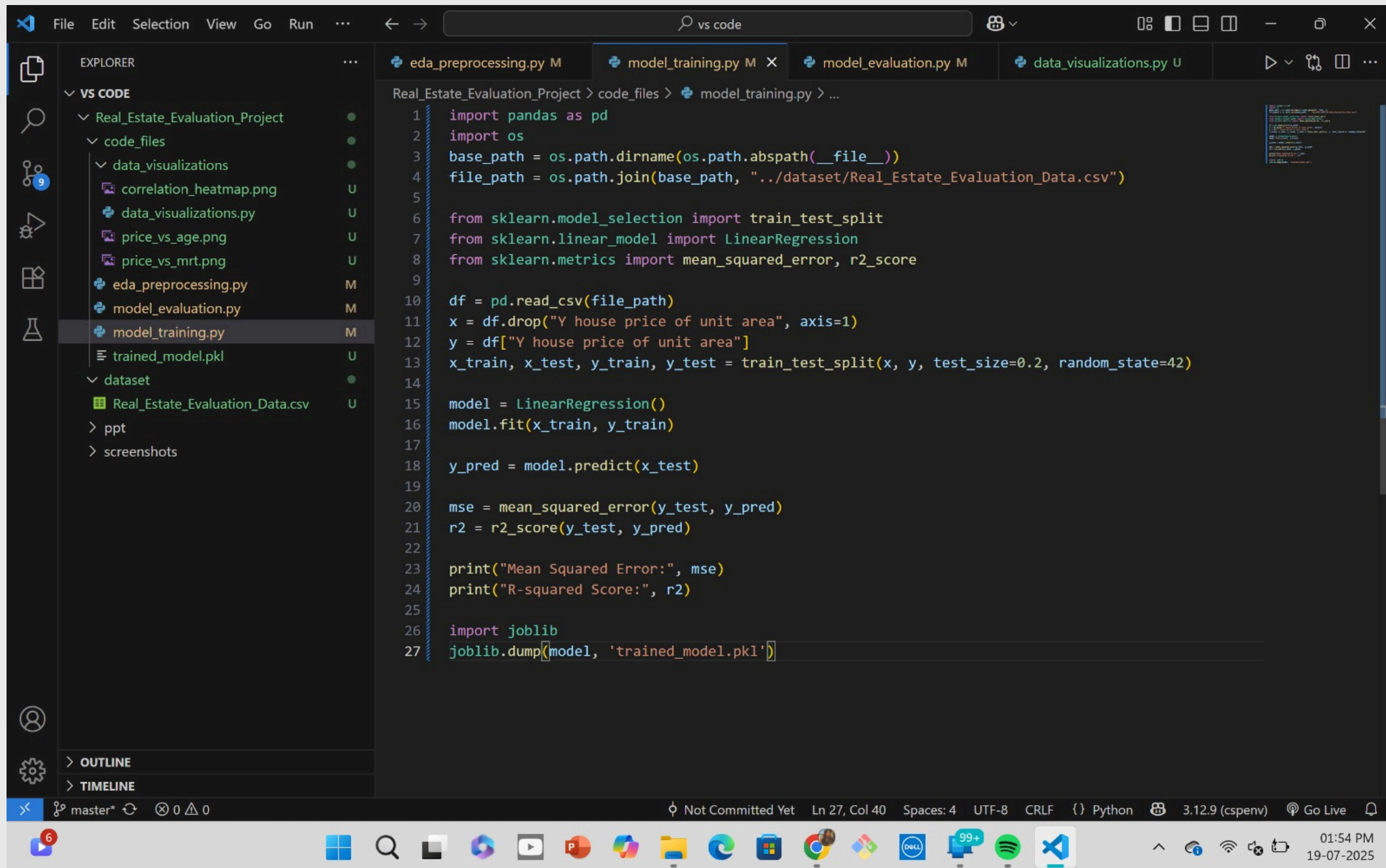


The screenshot displays the Visual Studio Code (VS Code) interface. On the left, the Explorer sidebar shows a project structure for 'Real_Estate_Evaluation_Project'. The file 'data_visualizations.py' is selected. The main editor area shows the 'TERMINAL' tab with the following output:

```
5 36.3 45.892396
6 53.0 44.338889
7 51.4 46.744485
8 16.1 24.432352
9 59.0 55.062585
Evaluation Results:
Mean Absolute Error (MAE): 5.418035098628801
Mean Squared Error (MSE): 54.60196067337408
Root Mean Squared Error (RMSE): 7.389313951468978
R-squared Score (R²): 0.6745228670351053
PS C:\vs code> & C:/Users/rishi/.conda/envs/cspenv/python.exe "c:/vs code/Real_Estate_Evaluation_Project/code_files/data_visualizations/data_visualizations.py"
```

The bottom status bar indicates the current file is 'data_visualizations.py' at line 15, column 33, using Python 3.12.9 (cspenv). The Windows taskbar at the very bottom shows the time as 02:07 PM on 19-07-2025.

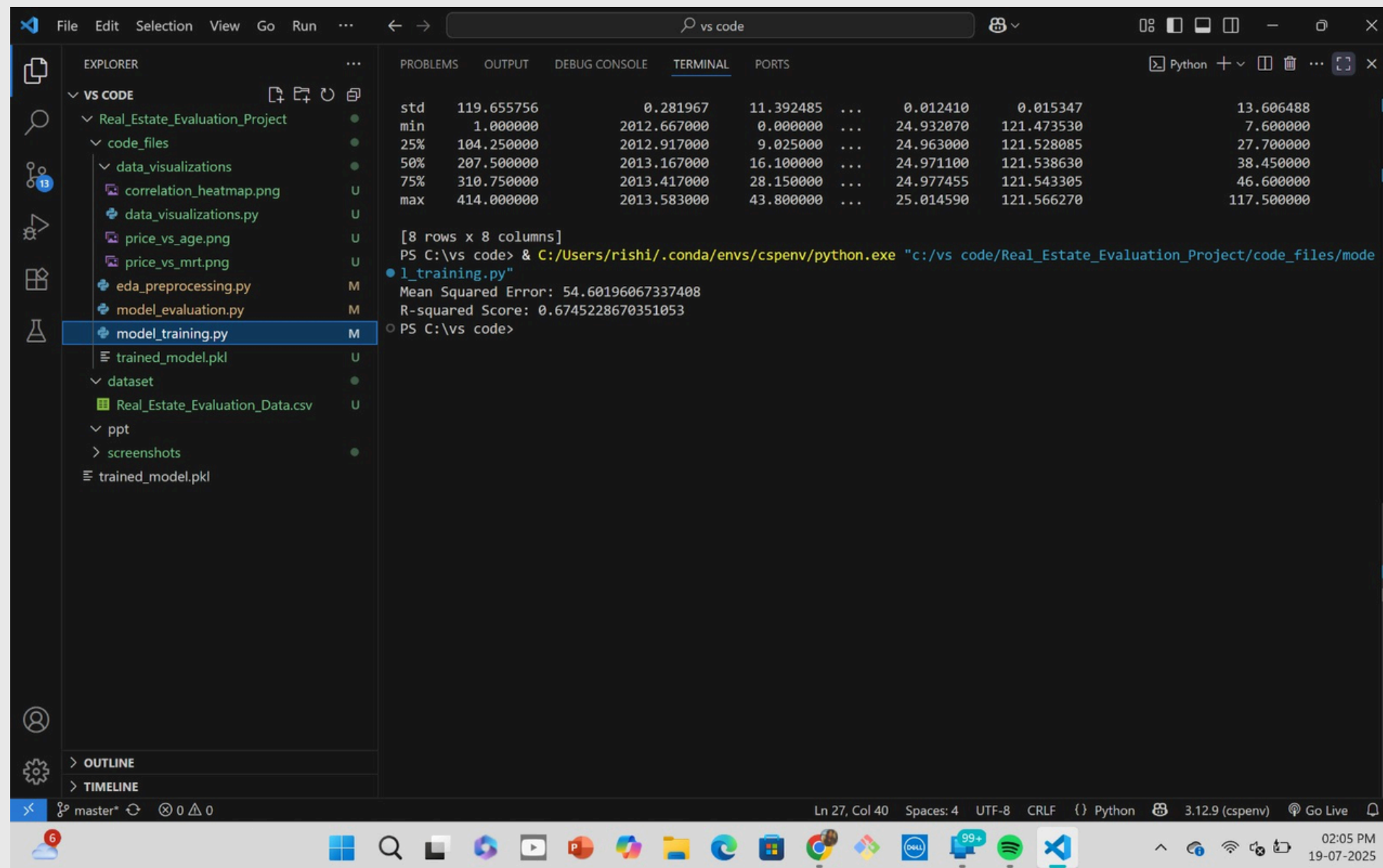
MODEL TRAINING



```
1 import pandas as pd
2 import os
3 base_path = os.path.dirname(os.path.abspath(__file__))
4 file_path = os.path.join(base_path, "../dataset/Real_Estate_Evaluation_Data.csv")
5
6 from sklearn.model_selection import train_test_split
7 from sklearn.linear_model import LinearRegression
8 from sklearn.metrics import mean_squared_error, r2_score
9
10 df = pd.read_csv(file_path)
11 x = df.drop("Y house price of unit area", axis=1)
12 y = df["Y house price of unit area"]
13 x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
14
15 model = LinearRegression()
16 model.fit(x_train, y_train)
17
18 y_pred = model.predict(x_test)
19
20 mse = mean_squared_error(y_test, y_pred)
21 r2 = r2_score(y_test, y_pred)
22
23 print("Mean Squared Error:", mse)
24 print("R-squared Score:", r2)
25
26 import joblib
27 joblib.dump(model, 'trained_model.pkl')
```



MODEL TRAINING OUTPUT



The screenshot shows the VS Code interface with the 'Real_Estate_Evaluation_Project' workspace. The 'code_files' directory contains several files, including 'model_training.py' which is currently selected. The terminal window displays the output of the training process, showing a summary of the dataset statistics and the results of the linear regression model.

```
std      119.655756      0.281967      11.392485      ...      0.012410      0.015347      13.606488
min      1.000000      2012.667000      0.000000      ...      24.932070      121.473530      7.600000
25%     104.250000      2012.917000      9.025000      ...      24.963000      121.528085      27.700000
50%     207.500000      2013.167000      16.100000      ...      24.971100      121.538630      38.450000
75%     310.750000      2013.417000      28.150000      ...      24.977455      121.543305      46.600000
max     414.000000      2013.583000      43.800000      ...      25.014590      121.566270      117.500000
```

[8 rows x 8 columns]
PS C:\vs code> & C:/Users/rishi/.conda/envs/cspenv/python.exe "c:/vs code/Real_Estate_Evaluation_Project/code_files/model_training.py"
Mean Squared Error: 54.60196067337408
R-squared Score: 0.6745228670351053
PS C:\vs code>

Text:

- Algorithm : Linear Regression
- Library : sklearn.linear_model
- Data split : 80% training, 20%testing
- Features : All 6 numerical columns
- Simple model, interpretable coefficients

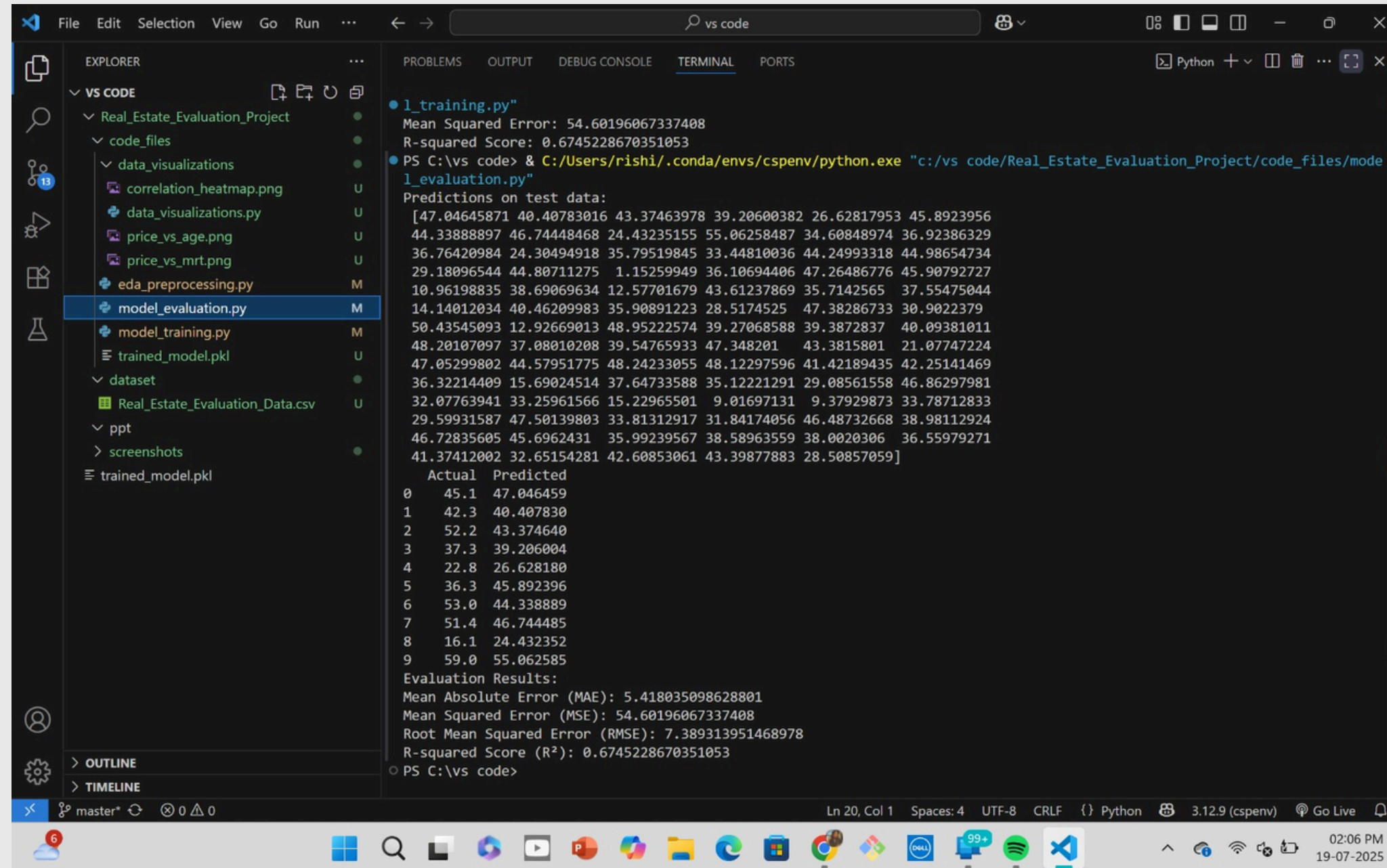
MODEL EVALUATION

```
1 import pandas as pd
2 import joblib
3 from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
4 import os
5 base_path = os.path.dirname(os.path.abspath(__file__))
6 file_path = os.path.join(base_path, "../dataset/Real_Estate_Evaluation_Data.csv")
7
8 df = pd.read_csv(file_path)
9 x = df.drop("Y house price of unit area", axis=1)
10 y = df["Y house price of unit area"]
11 from sklearn.model_selection import train_test_split
12 x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
13
14 model_path = os.path.join(base_path, "trained_model.pkl")
15 model = joblib.load(model_path)
16 y_pred = model.predict(x_test)
17 print("Predictions on test data:\n", y_pred)
18 comparison_df = pd.DataFrame({"Actual": y_test.values, "Predicted": y_pred})
19 print(comparison_df.head(10)) # Show first 10 comparisons
20
21
22 mae = mean_absolute_error(y_test, y_pred)
23 mse = mean_squared_error(y_test, y_pred)
24
25 import numpy as np
26 rmse = np.sqrt(mean_squared_error(y_test, y_pred))
27
28 r2 = r2_score(y_test, y_pred)
29
30 print("Evaluation Results:")
31 print("Mean Absolute Error (MAE):", mae)
32 print("Mean Squared Error (MSE):", mse)
33 print("Root Mean Squared Error (RMSE):", rmse)
34 print("R-squared Score (R²):", r2)
```

MODEL EVALUATION - OUTPUT

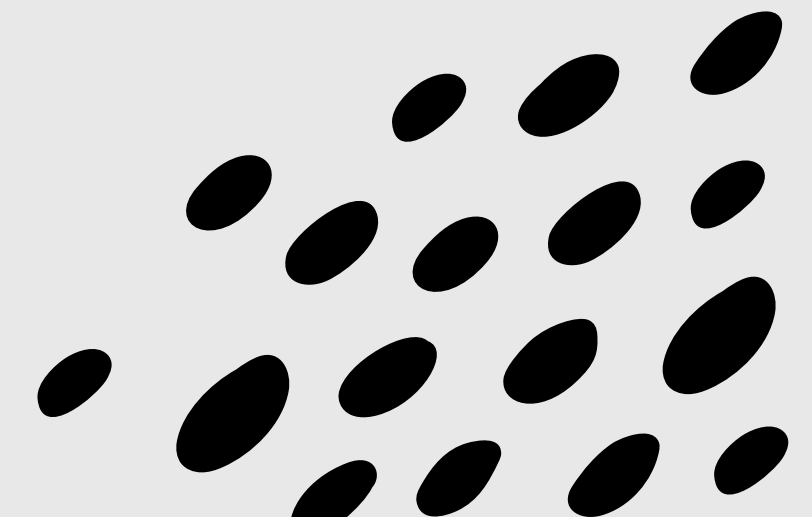
Text:

- Mean Absolute Error(MAE):
5.418035098628801
- Mean Square Error(MSE):
54.60196067337408
- R-square Score :
0.6745228670351053



The screenshot shows a VS Code interface with a terminal window open. The terminal displays the output of a Python script named `l_evaluation.py`. The output includes the Mean Squared Error (MSE), R-squared Score, and a table of predictions on test data. The evaluation results are as follows:

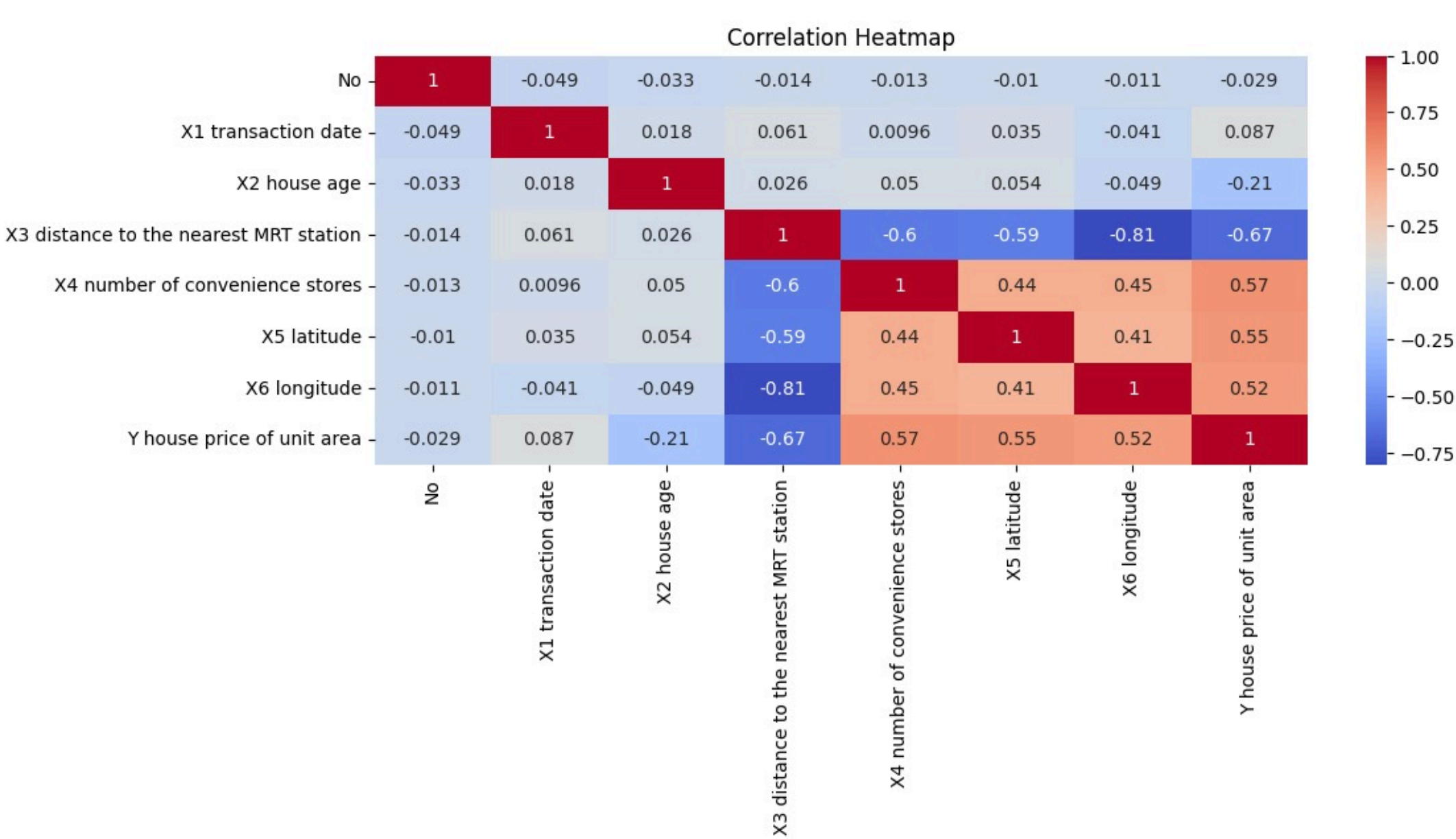
```
Mean Squared Error: 54.60196067337408
R-squared Score: 0.6745228670351053
Predictions on test data:
[47.04645871 40.40783016 43.37463978 39.20600382 26.62817953 45.8923956
 44.33888897 46.74448468 24.43235155 55.06258487 34.60848974 36.92386329
 36.76420984 24.30494918 35.79519845 33.44810036 44.24993318 44.98654734
 29.18096544 44.80711275 1.15259949 36.10694406 47.26486776 45.90792727
 10.96198835 38.69069634 12.57701679 43.61237869 35.7142565 37.55475044
 14.14012034 40.46209983 35.90891223 28.5174525 47.38286733 30.9022379
 50.43545093 12.92669013 48.95222574 39.27068588 39.3872837 40.09381011
 48.20107097 37.08010208 39.54765933 47.348201 43.3815801 21.07747224
 47.05299802 44.57951775 48.24233055 48.12297596 41.42189435 42.25141469
 36.32214409 15.69024514 37.64733588 35.12221291 29.08561558 46.86297981
 32.07763941 33.25961566 15.22965501 9.01697131 9.37929873 33.78712833
 29.59931587 47.50139803 33.81312917 31.84174056 46.48732668 38.98112924
 46.72835605 45.6962431 35.99239567 38.58963559 38.0020306 36.55979271
 41.37412002 32.65154281 42.60853061 43.39877883 28.50857059]
Actual Predicted
0 45.1 47.046459
1 42.3 40.407830
2 52.2 43.374640
3 37.3 39.206004
4 22.8 26.628180
5 36.3 45.892396
6 53.0 44.338889
7 51.4 46.744485
8 16.1 24.432352
9 59.0 55.062585
Evaluation Results:
Mean Absolute Error (MAE): 5.418035098628801
Mean Squared Error (MSE): 54.60196067337408
Root Mean Squared Error (RMSE): 7.389313951468978
R-squared Score (R²): 0.6745228670351053
```



CORRELATION HEATMAP(EDA)

Title : Feature Correlation withTarget

TEXT:



MRT distance: Strong negative correlation with price

Latitude: Moderate positive correlation

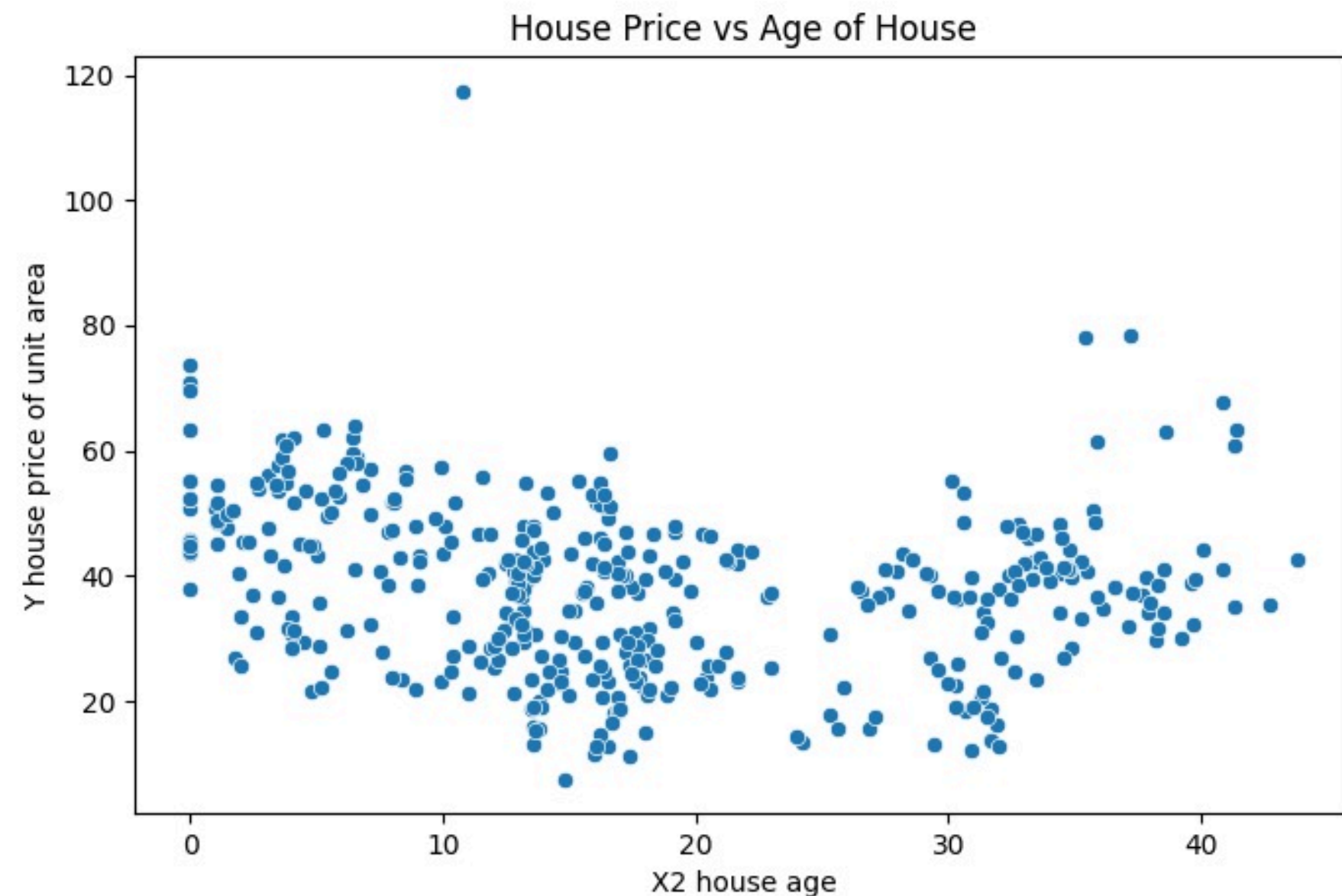
House age: Weak negative impact

HOUSE PRICE VS HOUSE AGE

Title : House price vs Age of House

Text:

- Newer house generally have higher prices
- No very strong trend

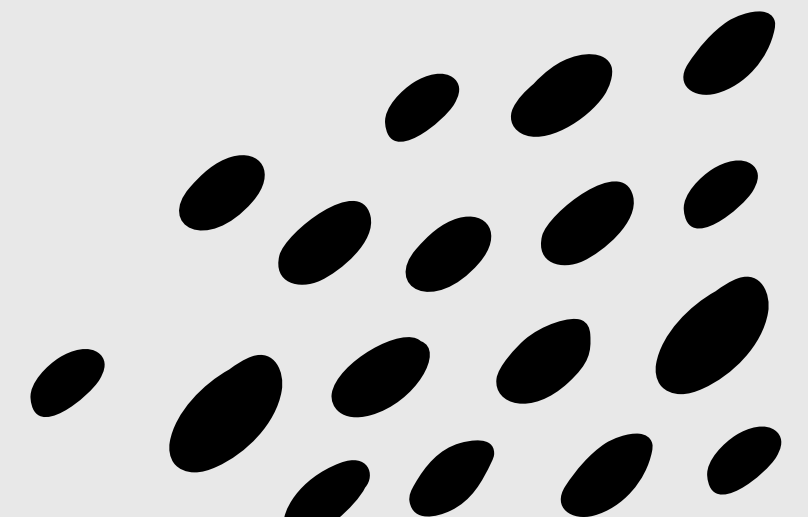
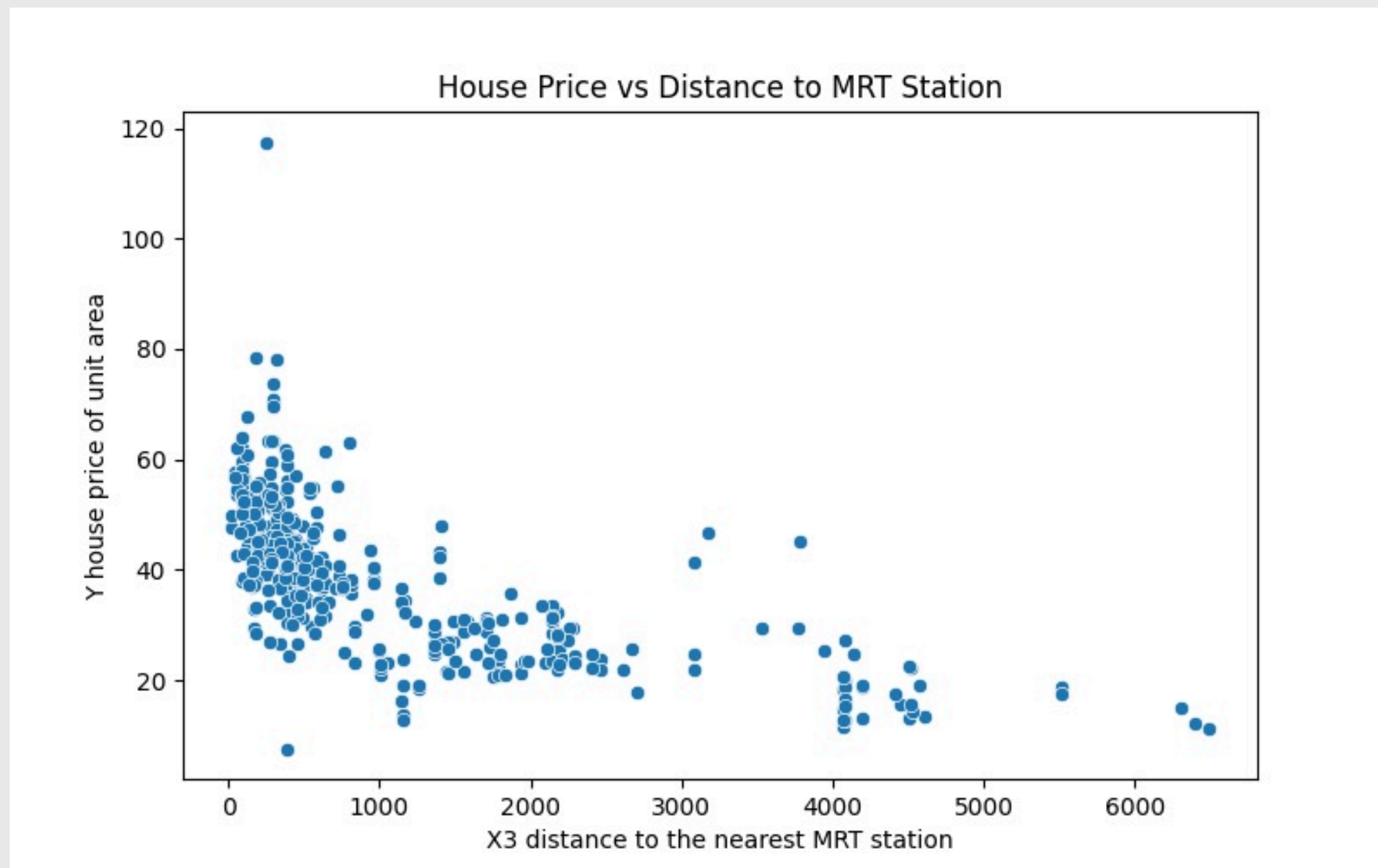


HOUSE PRICE VS DISTANCE TO MRT

Title : Impact of MRT Proximity on Price

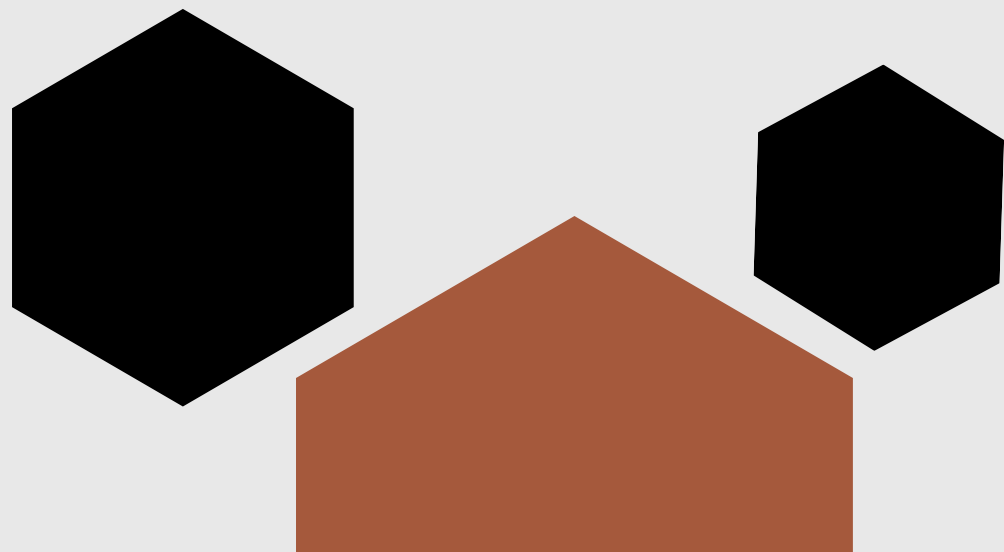
Text:

- Prices are higher near MRT stations
- Steep decline in price as distance increases



CONCLUSION

- Linear Regression works well with clear numerical data
- MRT distance is the strongest factor affecting price
- Improvements: Try advanced models(Random Forest,XGBoost),scaling,or adding time-based features



REFERENCES/ACKNOWLEDGEMENTS

Text:

- **Dataset** : UCI Machine Learning Repository
- **Tools**: Python, Pandas, Numpy, Matplotlib, seaborn,scikit-learn
- **IDE** : vs code
- **Group Members:**
 - Rishika V
 - Manasa J
 - Nur haliza H

